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Ex No 9 : Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objectives

Objective – 1: Analyze Channel Capacity (bps/Hz) Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB) Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

Objective – 3: Analyze Antenna Gain (dB) Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

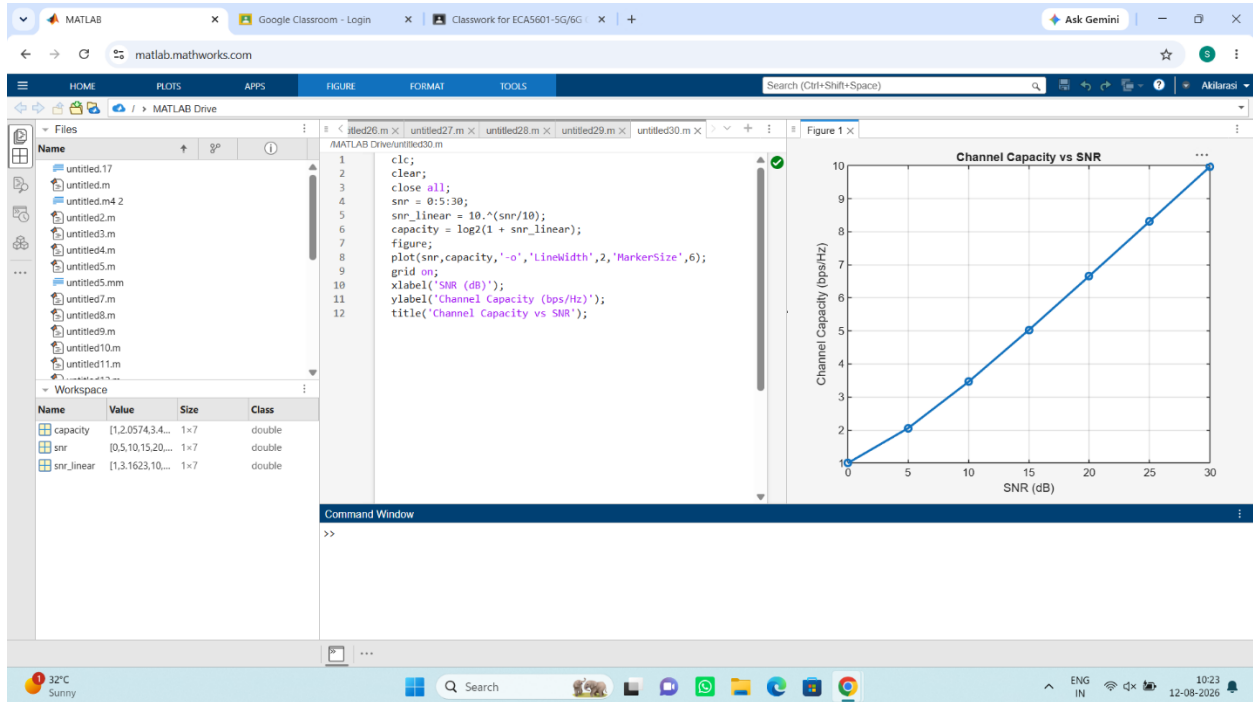
Procedure:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
```

```
ylabel('Channel Capacity (bps/Hz)');
```

```
title('Channel Capacity vs SNR');
```

Output:



Objective 2:

```
clc;
```

```
clear;
```

```
close all;
```

```
snr = 0:5:30;
```

```
snr_linear = 10.^(snr/10);
```

```
capacity = log2(1 + snr_linear);
```

```
figure;
```

```
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
```

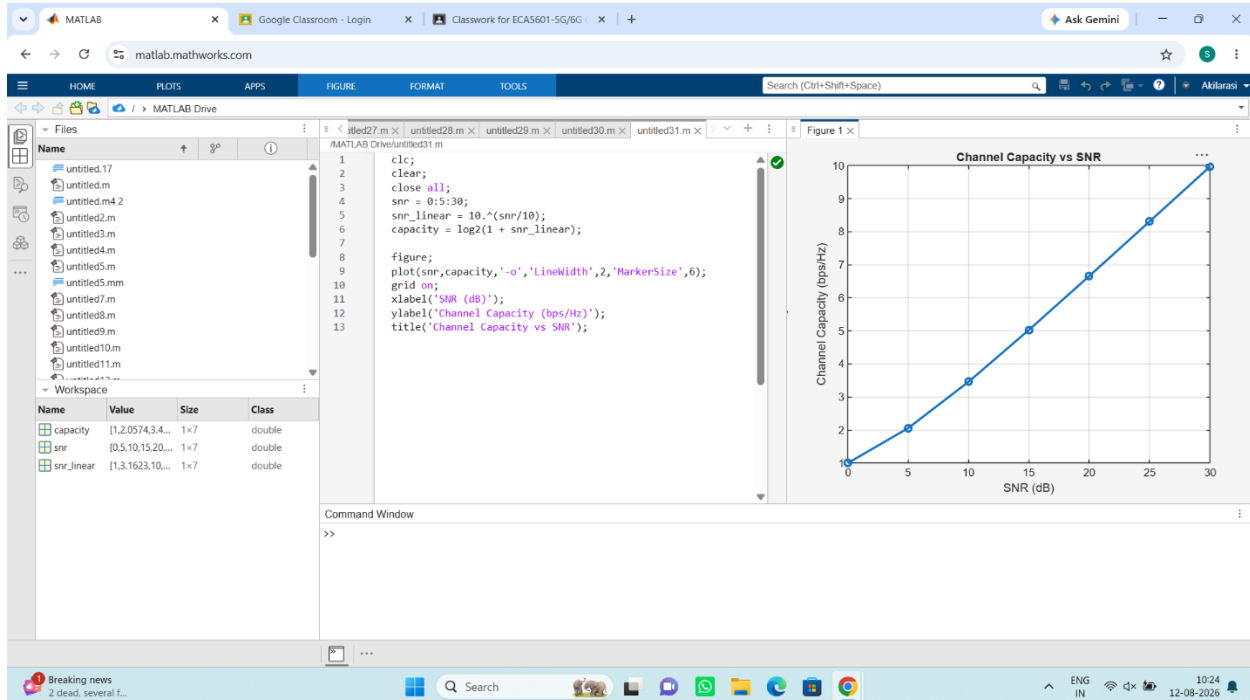
```
grid on;
```

```
xlabel('SNR (dB)');
```

```
ylabel('Channel Capacity (bps/Hz)');
```

```
title('192512216 Channel Capacity vs SNR');
```

Output:



Objective 3:

```
clc; clear; close all;
```

```
antennas = [4 8 16 32 64];
```

```
gain = 10*log10(antennas);
```

```
figure
```

```
bar(antennas, gain)
```

```
grid on
```

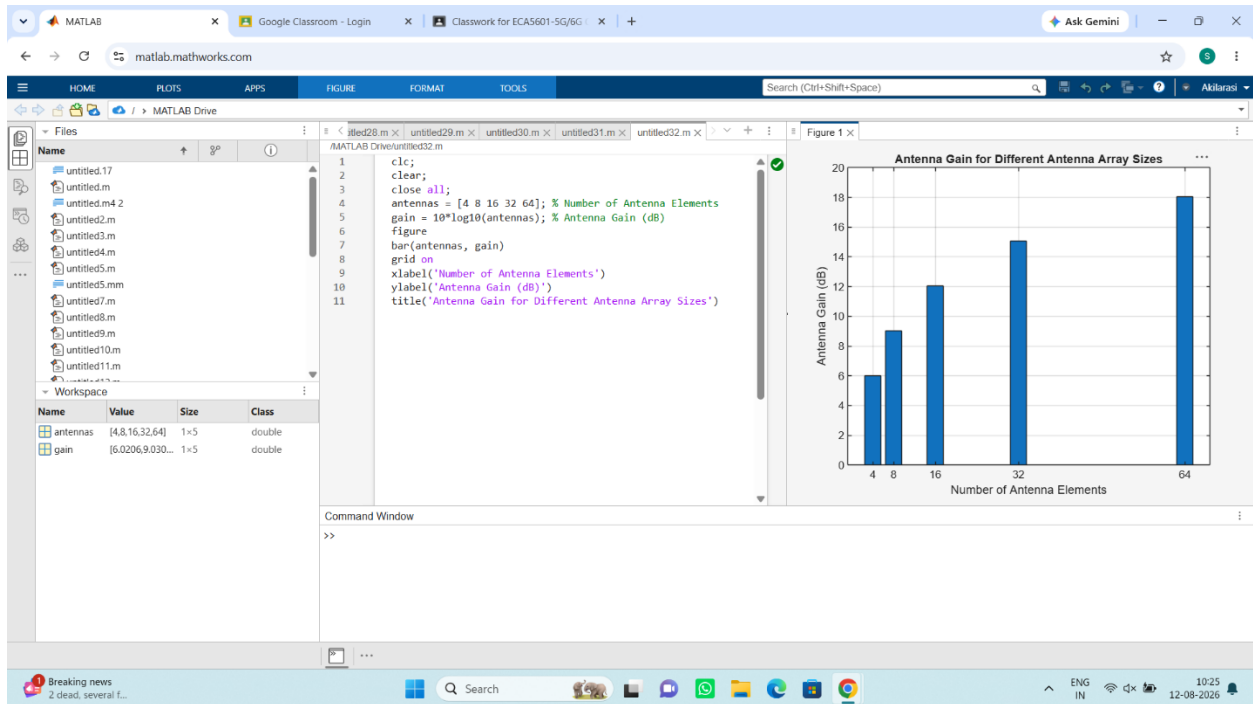
```
% Number of Antenna Elements % Antenna Gain (dB)
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Antenna Gain (dB)')
```

```
title('Antenna Gain for Different Antenna Array Sizes')
```

Output:



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.